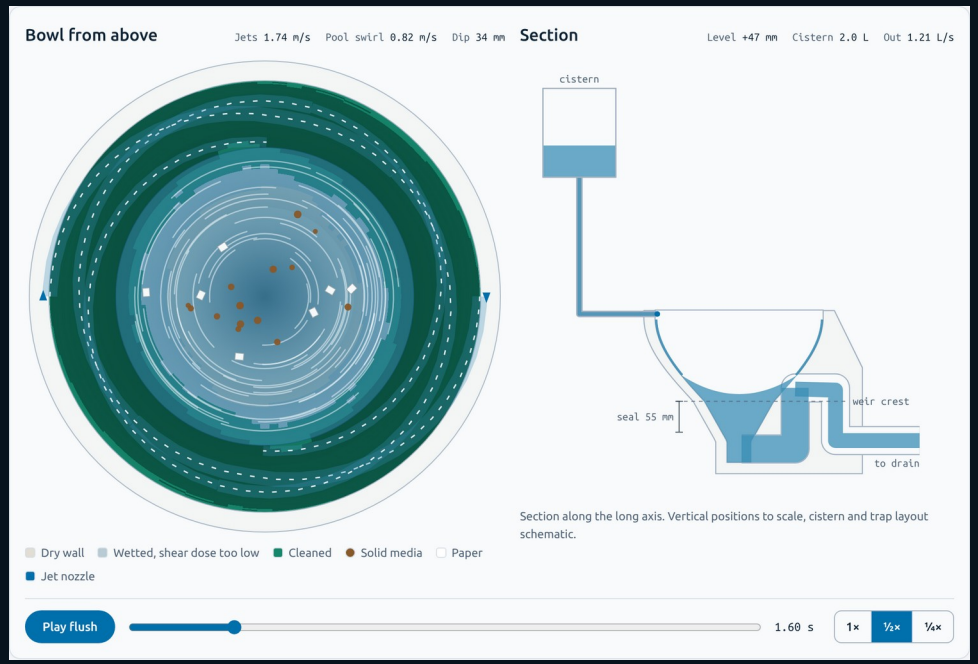
Tangential Jet Flush Model 2.0

A fast physics model of a rimless toilet flush, checked against CFD, with AI in the engineering loop



Flush playback from the page: films on the wall, pool swirl, waste. Washdown preset.

20–40 ms

per flush with the fast model, in the browser

≈ 2 hours

for 3.2 s of the same flush in OpenFOAM CFD, on 12 cores

4.5 vs 10.5

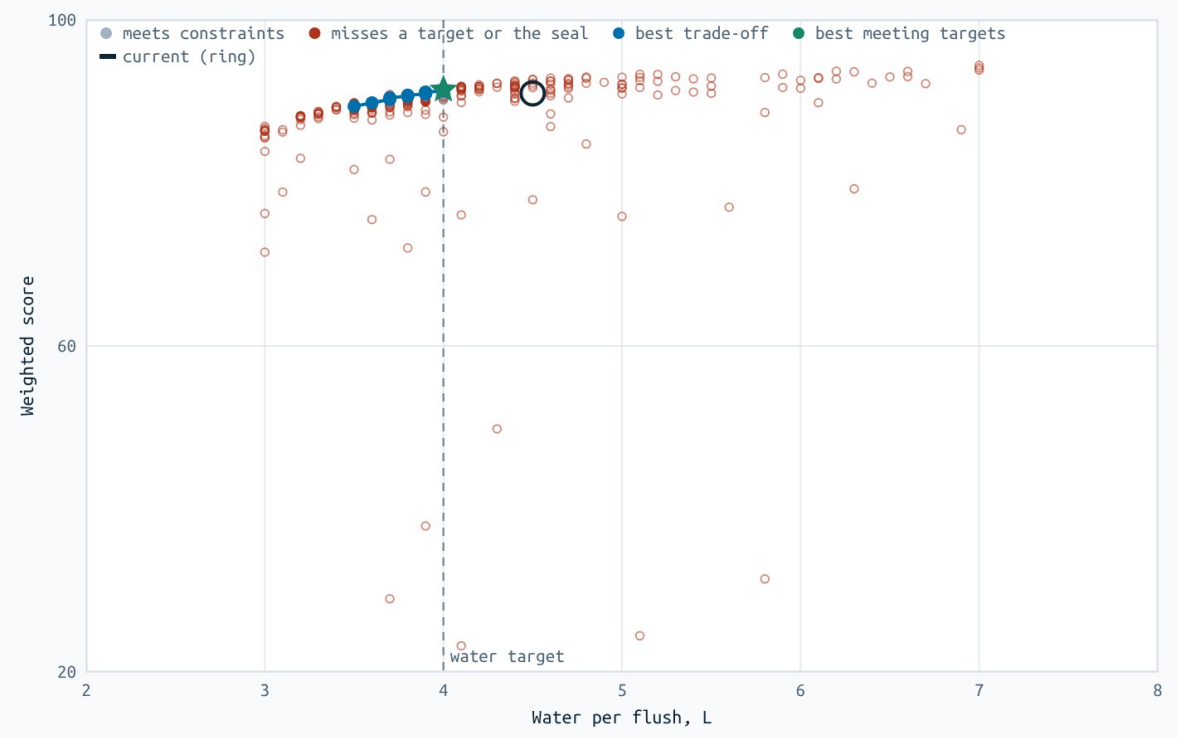
expensive runs to reach the targets, AI-assisted vs automated one change at a time (median of 8 synthetic projects)

Try it live fintlp.github.io/jet-flush-model

Reduced-order model, not yet calibrated or validated: results are for comparing designs, not measured values. The CFD is a second simulation of one design, not a measurement.

1 The fast model: one flush in milliseconds

Change the bowl or the jets and see cleaning, waste removal, water and noise at once; the AI lab searches for better designs.



AI lab: each dot is one simulated flush; the curve is the best trade-off between score and water.

From the guided demo, part 1

86 % → 79 %

cleaning when the bowl turns oval, jets at the ends of the long axis

→ 88 %

after moving the jets to the long sides

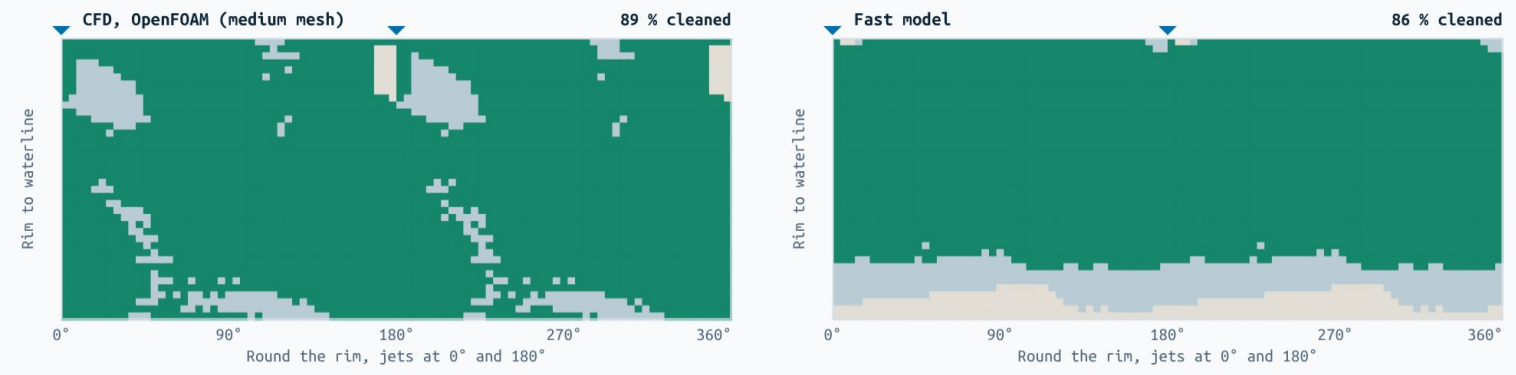
89 % at 4.0 L

AI lab design: 0.5 L less water than the oval start, −1.1 dB, no louder

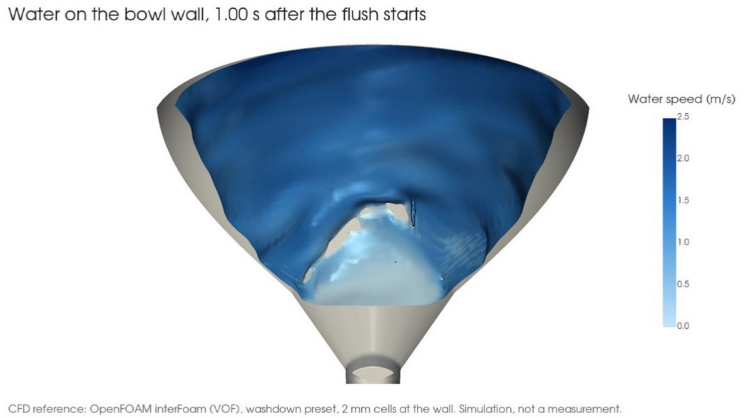
Illustrative estimates from the uncalibrated model; noise is a relative, experimental indicator. AI suggestions are candidates for a CFD or rig check, not final answers.

2 Checked against CFD

The same bowl and jets in OpenFOAM: about 2 hours on 12 cores for 3.2 s of flush, against about 30 ms for the fast model.



What gets cleaned, wall unrolled: CFD left, fast model right (green cleaned, grey wetted only).



CFD water surface at 1.0 s, front half removed.

89 % vs 86 %

of the wall cleaned, CFD vs fast model: close totals, different patterns

0.14 s vs 0.54 s

first water in the pool, CFD vs fast model: in the model it arrives later and all at once

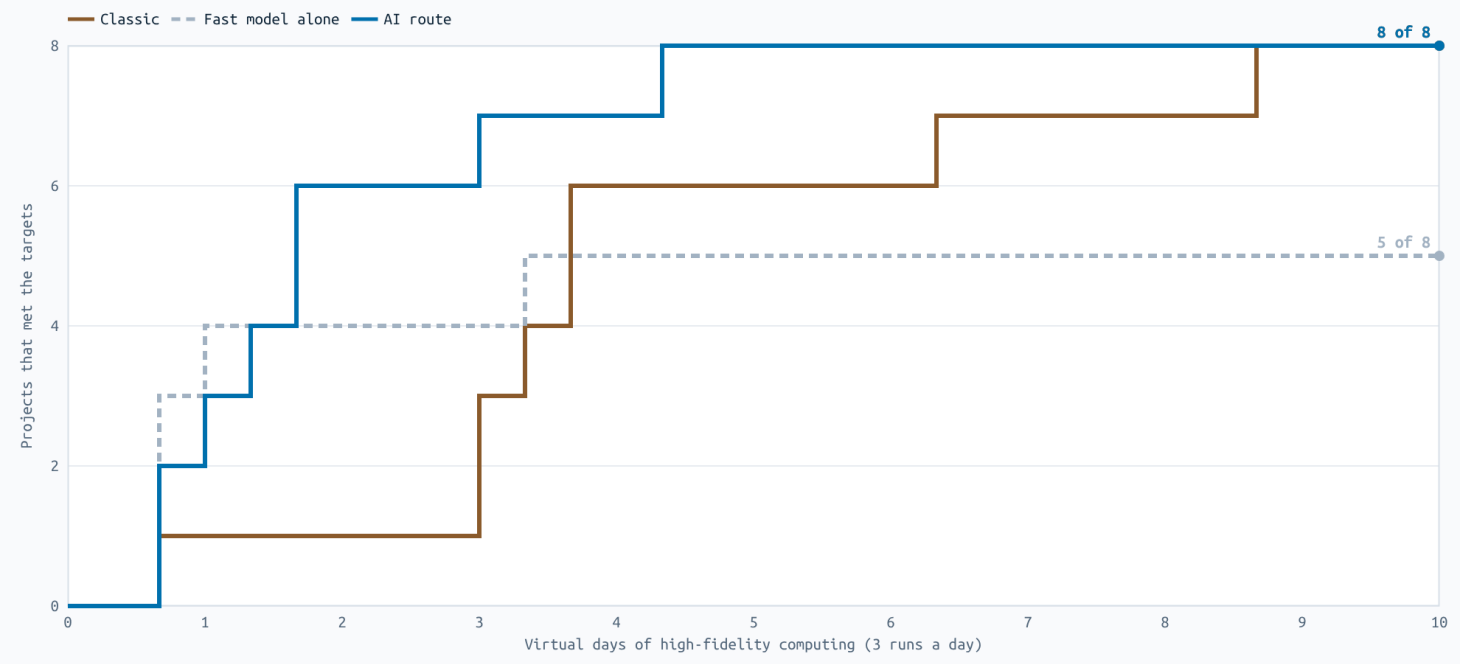
11 %

overlap of the uncleaned areas, CFD vs fast model: the model gets the total, not where the gaps are

A second simulation of one design, not a measurement or a validation. Trap outflow taken from the model; after 1.3 s the CFD outlet draws air. On two CFD meshes the totals agree, the uncleaned areas overlap 50 %: a sensitivity, not mesh independence.

3 AI in the engineering loop

Eight design projects, each run of the expensive reference costs 8 hours: how many runs until the targets are met?



Projects that met the targets over virtual days, for the three routes.

4.5 vs 10.5

expensive runs per project (median): fast model + learned correction vs automated one change at a time

43 vs 97

runs for all eight projects

3 of 8

projects the fast model alone never finished: it cannot learn from the expensive runs

The expensive runs come from a synthetic reference (the model with other coefficients and extra effects), labelled as such on the page. It shows the workflow, not real accuracy.

One design loop, two clocks

The median synthetic project of the challenge, counting each expensive iteration as a working day (set up, run, evaluate), as Geberit reported; the challenge itself counts 8 hours of computing per run.

One change at a time (automated)



10.5 days

AI-assisted



4.5 days

≈ 2.5 months

54 assumed working days of iterations, over eight synthetic projects

97 → 43

expensive iterations in total

Also in the AI lab, all in the browser

Design optimiser: about 540 quick estimates in about 10 s, then the recommended designs checked at full resolution

Neural surrogate: a few thousand times faster than the physics model

Hybrid calibration: fit plus a learned correction, tested on unseen designs

Noise calibration: fits the noise model to recordings (shown with synthetic ones)

Physics-informed network: finds a model coefficient from a level record

Field surrogate: whole cleaning maps from expensive runs, shown honestly

Iteration counts from the challenge against a synthetic reference (checks/challenge.js), not from real projects; one day per iteration is an assumption from the Geberit report by CADFEM. Sources on the page.

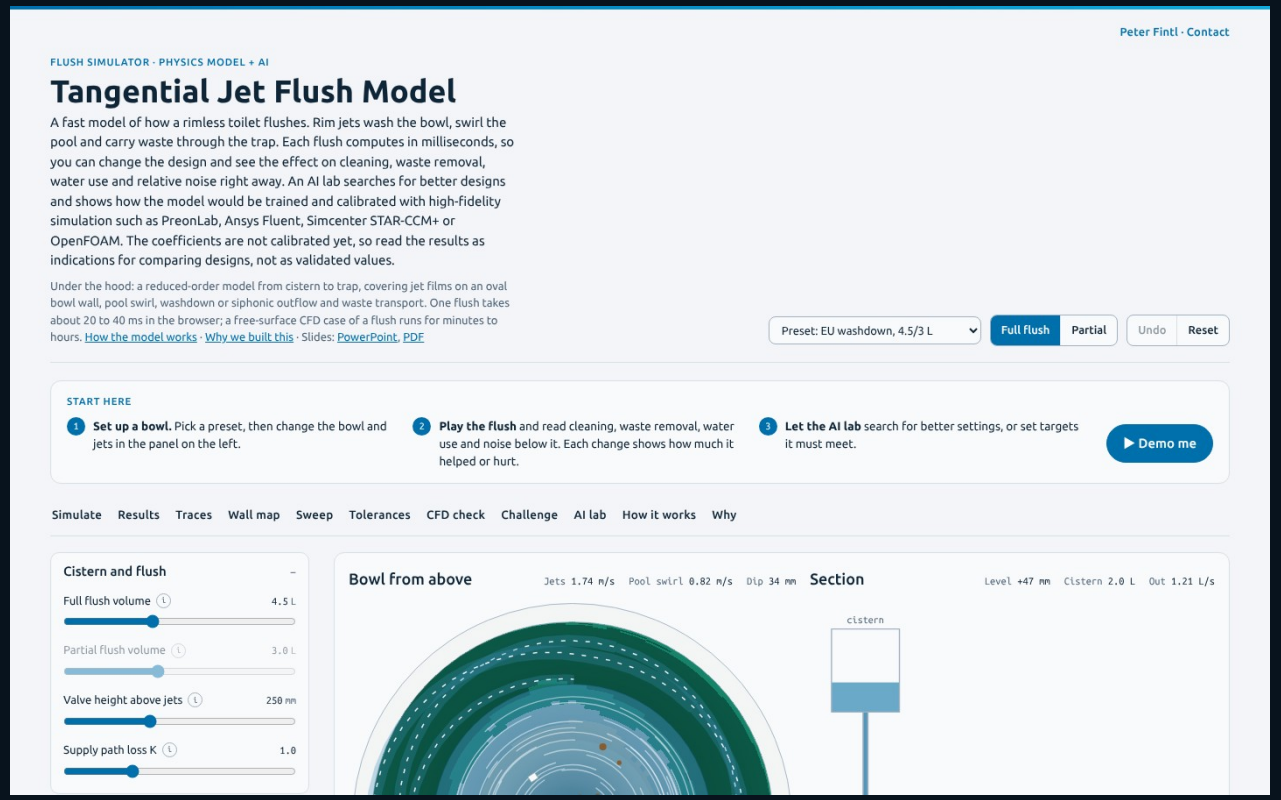
See the whole story

The guided story drives the live page in three parts. Each part can be started on its own.

1. **The fast model**
   fintlp.github.io/jet-flush-model/#demo

2. **Checked against CFD**
   fintlp.github.io/jet-flush-model/#cfd-tour

3. **AI in the engineering loop**
   fintlp.github.io/jet-flush-model/#challenge-tour



Recorded from the page (v2.0.0) on 21 October 2026: all three parts, at reading pace.

Physics model, not yet calibrated or validated. Synthetic references are labelled on the page; the CFD is a second simulation of one design, not a measurement.